Asrith Boppudi

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RESEARCH INTERESTS

- High-k dielectric scaling and physics-based compact modeling in sub-3nm FinFET and GAAFET architectures for leakage mitigation and power efficiency.
- Electromagnetic simulation and design of high-frequency integrated circuits, focusing on the integration and thermal control of Wide-Bandgap (GaN/SiC) devices.
- Bridging theoretical solid-state device physics with circuit-level performance by extracting layout parasitics and developing process-aware compact SPICE models.
- Thermo-mechanical stress modeling and signal integrity analysis in 3D-IC and System-in-Package (SiP) architectures, particularly for high-frequency communication systems.
- Low-power circuit design and system-level validation to meet stringent Size, Weight, and Power (SWaP) constraints for next-generation sensing and communication deployments.

EDUCATION

Bachelor of Technology in Electronics and Communication Engineering — Dec'21-May'25
Jawaharlal Nehru Technological University (MLRIT Affiliation), Hyderabad
NIRF Rank: 88
CGPA: 8.11 (First class with distinction)
Relevant Coursework: Engineering Physics, VLSI, Analog Electronic Devices, Digital System Design, RF and Microwave systems
Projects:

a. Design, Simulation and Fabrication of Hexagonal MIMO Patch Antenna for X-Band and UWB applications.
b. Cube satellite development for weather monitoring.
c. Design and manufacturing of ornithopter, for surveillance.
d. RISC V CPU Core
e. 32-bit superscalar RISC-V CPU
f. Smart IOT blind stick.

ONGOING RESEARCH PROJECT:

"**Impact of High-k Interfacial Layer (IL) Scaling on Gate Leakage and DIBL in 14nm FinFETs",** under the guidance of Dr.V. Radhika Devi, Principal and dean of MLRIT (JNTUH affiliation).

- Performing simulations of device physics utilizing **Synopsys Sentaurus TCAD** and **nanoHUB** to evaluate the scalability of High-k gate stacks within 14nm FinFET architectures.
- Assessing the trade-offs associated with Interfacial Layer (IL) thickness and **Drain-Induced Barrier Lowering (DIBL)** to enhance the control of Short Channel Effects (SCE).
- Analyzing the relationship between dielectric scaling and gate leakage mechanisms to recommend optimized device parameters for low-power logic nodes.

RESEARCH CONFERENCES ATTENDED:

- **International Conference on Graphene 2026.**
- **Outcome: Abstract paper accepted for presentation. Presented a comparative meta-analysis titled, *"Mitigation of Short Channel Effects Beyond Silicon: A Meta-analysis of 14nm FinFETs and 2D MoS2 Channel Architectures".***

PUBLICATIONS AND RESEARCH WORK

- 1.“Patent/IPO on **GROUND PLATFORM** for microgravity research.” (Under preparation, final stage) (Initiated: January 2025)
- This project involves the design and development of a clinostat for microgravity simulation on earth. The number of people involved in this project are 6. My role in this patent is to design the entire electronic architecture and power requirements for the device. Expected to be published by May 2026.
- 2. “Research paper on **LAB-ON-CHIP** for cancer diagnostics.” (Under preparation, final stage) (Initiated: May 2025)
- This project involves the design and development of a microfluidics chip for cancer diagnostics. The number of people involved in this project are 3. My role in this research project was to simulate the **deposition of HfO2 thin films** using TCAD tools, by analysing the impacts of dielectric parameters on LOC. Expected to be published by April 2026.
- 3. “Design, simulation and fabrication of Hexagonal MIMO patch antenna.” (Submitted for IEEE CONECCT, under review) (Initiated: May 2025).

PROFESSIONAL EXPERIENCE

Electronic System Designer (full time) (Pre-Placement) Apr’24-Oct’25

Department of Avionics, Resolute Lab India, Bareilly

- Built high-density mixed-signal boards in Altium Designer and KiCAD, optimizing System-in-Package (SiP) floorplanning and signal routing to minimize parasitic capacitance for high-performance hardware payloads.
- Integrated ultra-low-noise sensors for Lab-on-Chip systems via I2C/SPI protocols, carefully tailored Power Delivery Networks (PDN) to reduce leakage currents and adhere strictly to low-power (SWaP) constraints.
- Designed a 4x4 RF antenna array system, conducting advanced electromagnetic simulations in CST Microwave Studio Suite for beamforming validation, impedance matching, and high-frequency RF link budget optimization.
- Assessed system-level power dissipation and thermo-mechanical stress from extreme thermal cycling; skills that are directly applicable to advanced 3D-IC packaging and thermal management. Developed HIL-validated electronic architectures with finely tuned PID control loops.
- Leveraged standard IC communication protocols (CAN, UART) for precise hardware synchronization, conducting extensive validation to fulfill ISRO's rigorous Quality Assurance and "Zero Defect" manufacturing reliability standards.

PROFESSIONAL TRAINING EXPERIENCE

1. **Digital Design Verification Trainee** Oct’25-Present
   VLSI For ALL Pvt Ltd, Uttar Pradesh, India
   - Verilog
   - System Verilog

2. **Trainee**
   Indian Semiconductor Workforce Development Jun’25-Sep’25
   (Samsung, IISc Bangalore, Synopsys)

   - Gained hand-on experience on Synopsys Sentaurus TCAD to analyze and simulate the semiconductor process steps and simulation, especially on transistors like FinFET and MOSFETs.
   - Analyzed the impact of doping, gate oxide profiles, and recombination mechanisms in sub-micron regimes by simulations.
   - Analyzed the impact of High K dielectrics on gate leakages.

INTERNSHIPS:

1. UR Rao Satellite Centre, Indian Space Research Organisation (ISRO) Nov’24- Dec’24

   - Contributed to simulation and analysis of analog and digital sub-circuits utilizing PSpice/LTspice, thereby validating theoretical principles in relation to practical design limitations.
   - Conducted comparative analyses of the electronic components, such as operational amplifiers and regulators, by examining the datasheet specifications to recommend appropriate parts for high-reliability applications.
   - Acquired practical experience on Space-Grade (High-Rel) design methodologies, comprehending the effects of radiation and thermal extremes on the performance of electronic circuits.

2. Resolute Lab India Oct’23-Mar’24

   - Performed R&D on sensors.
   - Researched and developed a Thermal imaging sensor for imagery.

3. Brahmastra Aerospace Pvt Ltd Sep’23-Oct’23

   - Designed and Simulated Helical antenna.

- Conducted research on materials and manufacturing processes to support project development.
- Worked on signal processing using ML.

ACADEMIC PROJECTS

1. **Designed and fabricated a MIMO patch antenna for X-Band and UWB applications (Major)** **Aug'24-May'25**
   - The key outcome is, the antenna is successfully designed and fabricated. It has a desirable return loss of -38dB.
   - This project involves the design and fabrication of strip antenna for UWB applications.
   - The number of people involved are 4. My role in this project was to design, math modelling, and fabrication of the antenna.

2. **32-bit superscalar RISC-V CPU. (Personal Project under HoD)** **Aug'24-Nov'24**
   - Developed a UVM-based verification environment for a Superscalar RISC-V core, which involved creating agents to drive random instruction streams and monitor the validity of dual-issue operations.
   - Implemented functional coverage models aimed at verifying corner cases, with a specific focus on "back-to-back" dependency hazards and ALU-to-ALU forwarding paths within parallel pipelines.
   - Conducted comprehensive regression testing to validate the Instruction Issue Logic, ensuring that the processor accurately manages the stalling or splitting of issue packets when hardware resources or dependencies hinder dual execution.

3. **RISC V CPU Core. (Personal university project)** **Feb'24-May'24**
   - Developed a 32-bit Single-Cycle RISC-V Core that implements the RV32I instruction set, with the datapath and control unit meticulously designed from the ground up in Verilog.
   - Conducted functional verification through the creation of directed testbenches in System Verilog, effectively validating the arithmetic, logical, load/store, and branch instructions
   - Synthesized and analyzed the design in accordance with timing and area constraints, optimizing the critical path delay to enhance the maximum operating frequency.
4. **Designed and developed a 1U cube satellite using Arduino Nano. (Mini project) (O grade)** **Nov'23-Jan'24**
   - The key outcome of this project is, the satellite received signals successfully from the transmitter with desirable values.
   - I am responsible in integrating the sensors with the MCU.

5. **Developed an ornithopter drone for surveillance. (Personal university project)** **Mar'23-Jul'23**
   - Achievement: Special recognition from Head of the Department.
   - The key outcome of this project is, the drone gliding was successful, and it captured the data within the altitude range of 1.5km.
   - I led the electronics integration for the UAV by managing the link budget for RC communication and battery optimization for the higher altitudes.

TECHNICAL SKILLS

1. Hardware Descriptive Language: Verilog and System Verilog, UVM (under training)
2. Cadence Xcelium and Modelsim: For VLSI digital IC design verification
3. SPICE modelling (LTSpice,PSpice)
4. KiCad/Proteus software: For PCB design and Simulation
5. Ansys HFSS/CST: For antenna design and simulation.
6. TCAD Synopsys Sentauraus, nanoHUB muGFET , DEVSIM python scripting and COMSOL Multiphysics: For device physics simulation.

PROFESSIONAL CERTIFICATIONS

1. **ISWDP (Indian Semiconductor Workforce Development Program from Samsung, IISc Bangalore, Synopsys):** Level-1 (90%) (Beginner), Level-2 (76%) (Intermediate) and Level-3 (82%) (Advanced) **Jun'25-Sep'25**
2. **NPTEL on ELECTRONIC SYSTEM DESIGN from IIT DELHI:** Elite Silver **Jul'24-Oct'24**
3. **NPTEL course on RF and MICROWAVE from IIT KHARAGPUR:** Elite Bronze **Jan'24-Apr'24**
4. **PCB design from Internshala.** **Apr'23-May'23**

ONGOING CERTIFICATIONS

1. Semiconductor Devices for Next Generation Field Effect Transistors (More than Moore): A Physics Perspective (IIT Roorkee) **Jan'26-Apr'26**

AWARDS AND HONORS

1. Top 20% in college academics. (One of the highest grades in the department).